



RS6241

Hardware Design and PCB Layout

User Guide

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1 Features

- **FMCW mmWave Radar**
 - 57-66 GHz coverage
 - Two transmit channels and four receive channels
 - Integrated RFPLL, DPLL, transmitter, receiver, analog-to-digital conversion, and baseband
 - 12.0 dBm typical transmitter output power
 - 10.5 dB typical receiver noise figure
 - -95 dBc/Hz at 1 MHz typical phase noise
 - 10/20 MHz IF bandwidth
 - Sweep rate up to 800 MHz/μs
 - HWA for 1D/2D FFT, static-clutter removal, and CFAR operations
- **2.4 GHz Wireless Connectivity**
 - Protocols: BLE 5.3, BLE Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and proprietary 2.4 GHz protocols
 - Bluetooth PHY: 1 Mbps, 2 Mbps, Long Range S2 (500 kbps), and S8 (125 kbps)
 - Maximum TX output power: 10 dBm in BLE mode
 - RX sensitivity: -95 dBm at BLE 1 Mbps
- **Application System**
 - Dual-core 32-bit RISC-V CPU with DSP, FPU (up to 256 MHz), and I/D caches for radar post-processing, wireless connectivity, and applications
 - Security engine: secure boot, eFuse key protection, dynamic SRAM and XIP decryption, TRNG, AES-128, SHA, RSA, and ECC
- **Memory**
 - 551 KB on-chip SRAM
 - External PSRAM expansion up to 16 MB
- **Host and Device Interfaces**
 - 3 UART interfaces
 - 2 QSPI interfaces with XIP
 - X2 I²C
 - 8 PWM channels
 - X10 GPADC
 - 1 LIN controller
 - 1 CAN-FD controller
- **Clock Sources**
 - 32/40 MHz crystal oscillator as the clock reference
 - Internal LSI and MSI RC oscillators
 - Optional 32 kHz crystal for low-power operation
- **Power Management**
 - 1.7 V to 5.5 V single supply
 - Built-in LDO network for enhanced PSRR
 - BOM-optimized and power-optimized modes
- **Ultra-Low-Power Features**
 - Power states: off, standby, idle, and active (sensing, processing, and connectivity)
 - Fast wake-up from standby to active
- **Package**
 - FCCSP: 6.7 x 6.7mm² 94 pins
- **Temperature Range**
 - Operating junction temperature: -40 °C to 125°C

2 Pin Descriptions

2.1 Chip Pin Diagram

TOP VIEW

	1	2	3	4	5	6	7	8	9	10
A	GND	ANT	GND	TX1	GND	TX0	GND	GND	VRF	GND
B	GND	GND	GND		GND		GND	GND	GND	GND
C	XL2	GND	GND	GND	GND	GND	V13T	GND		RX0
D	XL1	GND	GND	V13M	GND	GND	GND	GND	GND	GND
E	PA12	GND	GND	V18C	GND	GND	GND	GND		RX1
F	PA11	PA13	PA14	PA15	V18A	V13R	GND	GND	GND	GND
G	PA2	PA3	PA8	PA9	PA10	PA7	GND	GND		RX2
H	VDDIO	PA1	PA4	PA5	PA6	PA0	NRST	GND	GND	GND
K	VEXT	VCC	VINT	PB0	PB5	PB3	PB2	GND		RX3
L	GND	VCC	VDIG	PB1	PB4	PB6	PB7	GND	GND	GND

Figure 2-1 Chip Pin Diagram

2.2 Pin Descriptions

Table 2-1 Pin Descriptions

Pin Name	I/O	Pin No.	Pin Description
VCC	Power In	K2,L2	Power input, 1.7~5.5V
VINT	Power Out	K3	INT LDO Output, 1.8 V by default, 0.9 to 2.5 V in 0.1 V steps
VEXT	Power Out	K1	EXT LDO Output, 3.3 V by default, 1.6 to 3.3 V in 0.1 V steps
VDIG	Power Out	L3	DIG LDO Output, 1.15 V by default, 0.65 to 1.35 V in 0.05 V steps
VDDIO	Power In	H1	GPIO Power input, 1.7~3.6V
V18A	Power In	F5	MMWAnalog power input, 1.5~2.5V
V18C	Power In	E4	mmWave analog power input, 1.5 to 2.5 V
V13T	Power In	C7	mmWave RF transmit power input, 1.2 to 2.5 V
V13R	Power In	F6	mmWave RF receive power input, 1.2 to 2.5 V
VRF	Power Out	A9	mmWave RF LDO output, 1.0 V by default
V13M	Power In	D4	2.4G Analog power input, 1.2 to 2.5 V
XL1	Analog	D1	Crystal oscillator output
XL2	Analog	C1	Crystal oscillator input
ANT	RF	A2	2.4GAntenna Port

TX0	RF	A6	Radar antenna output 0
TX1	RF	A4	Radar antenna output 1
RX0	RF	C10	Radar antenna input 0
RX1	RF	E10	Radar antenna input 1
RX2	RF	G10	Radar antenna input 2
RX3	RF	K10	Radar antenna input 3
GND	Ground	/	Chip ground
NRST	In	H7	Chip reset; pulled up to VDDIO by default
PA0	In/Out	H6	GPIO input/output, Wakeup IO
PA1	In/Out	H2	GPIO input/output, Wakeup IO
PA2	In/Out	G1	GPIO input/output, Wakeup IO
PA3	In/Out	G2	GPIO input/output, Wakeup IO
PA4	In/Out	H3	GPIO input/output, Wakeup IO
PA5	In/Out	H4	GPIO input/output, Wakeup IO
PA6	In/Out	H5	GPIO input/output, Wakeup IO
PA7	In/Out	G6	GPIO input/output, Wakeup IO
PA8	In/Out	G3	GPIO input/output, Wakeup IO
PA9	In/Out	G4	GPIO input/output, Wakeup IO
PA10	In/Out	G5	GPIO input/output, Wakeup IO
PA11	In/Out	F1	GPIO input/output, Wakeup IO
PA12	In/Out	E1	GPIO input/output, Wakeup IO
PA13	In/Out	F2	GPIO input/output, Wakeup IO
PA14	In/Out	F3	GPIO input/output, Wakeup IO
PA15	In/Out	F4	GPIO input/output, Wakeup IO
PB0	In/Out	K4	GPIO input/output, Wakeup IO
PB1	In/Out	L4	GPIO input/output, Wakeup IO
PB2	In/Out	K7	GPIO input/output
PB3	In/Out	K6	GPIO input/output
PB4	In/Out	L5	GPIO input/output
PB5	In/Out	K5	GPIO input/output
PB6	In/Out	L6	GPIO input/output
PB7	In/Out	L7	GPIO input/output

2.3 GPIO Multiplexing Table

Table 2-2 GPIO Multiplexing Table

GPIO Name	Func2	Func3	Func4	Func5	Func6	Func7	Func after rst
-----------	-------	-------	-------	-------	-------	-------	-------------------

PA0/WIO0	SPI0_CS	UART1_TX	CANFD_TX		PWM0		OFF/No-Pull
PA1/WIO1	SPI0_CLK	UART1_RX	CANFD_RX		PWM1		OFF/No-Pull
PA2/WIO2	SPI0_MOSI	UART1_RTS	I2C0_SDA	UART2_TX/LIN_TX	PWM2		OFF/No-Pull
PA3/WIO3	SPI0_MISO	UART1_CTS	I2C0_SCL	UART2_RX/LIN_RX	PWM3		OFF/No-Pull
PA4/WIO4	SPI0_HOLD	I2C0_SDA			PWM4		OFF/No-Pull
PA5/WIO5	SPI0_WP	I2C0_SCL			PWM5		OFF/No-Pull
PA6/WIO6	UART1_RTS	UART2_TX	I2C1_SDA		PWM6	GPADC6	OFF/No-Pull
PA7/WIO7	UART1_CTS	UART2_RX	I2C1_SCL		PWM7	GPADC7	OFF/No-Pull
PA8/WIO8/LXL1 ¹	UART1_TX	I2C1_SCL			PWM0	GPADC0	OFF/No-Pull
PA9/WIO9/LXL2 ¹	UART1_RX	I2C1_SDA			PWM1	GPADC1	OFF/No-Pull
PA10/WIO10	I2C1_SDA	CANFD_TX			PWM2	GPADC2	OFF/No-Pull
PA11/WIO11	I2C1_SCL	CANFD_RX			PWM3	GPADC3	OFF/No-Pull
PA12/WIO12	UART2_RX	UART1_RTS			PWM4	GPADC4	OFF/No-Pull
PA13/WIO13/TEST ²	UART2_TX	MCO_CLK			PWM5	GPADC5	IN/UP
PA14/WIO14	TWIO_SDA	UART2_RX		CANFD_TX	PWM6		OFF/No-Pull
PA15/WIO15	TWIO_SCL	UART2_TX		CANFD_RX	PWM7		OFF/No-Pull
PB0/WIO16	UART0_TX				PWM0		OFF/UP
PB1/WIO17	UART0_RX				PWM1		OFF/UP
PB2	UART0_RTS	FC_WP	SPI0_WP		PWM2		OFF/UP
PB3	UART0_CTS	FC_HOLD	SPI0_HOLD		PWM3		OFF/UP
PB4	FC_CS	UART2_RX	SPI0_CS		PWM4		OFF/No-Pull
PB5	FC_CLK	UART2_TX	SPI0_CLK		PWM5		OFF/No-Pull
PB6	FC_MOSI	I2C0_SDA	SPI0_MOSI		PWM6		OFF/No-Pull
PB7	FC_MISO	I2C0_SCL	SPI0_MISO		PWM7		OFF/No-Pull

- (1) LXL1 and LXL2: These are the 32 kHz crystal pins multiplexed with PA8 and PA9. Disable the GPIO function when using the analog function.
- (2) After initial power-on or reset, PA13 is configured as a pulled-up input. If this I/O is pulled to GND at that time, the chip enters ATE test mode. Therefore, PA13 must not be pulled low by default.

3 Peripheral Interface Features

3.1 GPIO

RS6241 provides 24 general-purpose input/output (GPIO) pins. All ports are mapped through alternate-function multiplexing. GPIO functions can be multiplexed on multifunction I/O pins by selecting alternate functions in the GPIO controller registers.

- GPIO configuration features:
- Three modes: input, output, and high impedance
- Configurable input pull-up and pull-down
- Default output drive strength of 20 mA, configurable in 20 mA steps up to 80 mA
- I/O outputs support push-pull and open-drain modes
- External-interrupt I/Os support five trigger modes: rising edge, falling edge, both edges, high level, and low level
- The 18 wake-up I/Os can be configured to wake the system from standby through an external interrupt
- All I/Os can be configured to wake the system from idle through an external interrupt

Note:

Eight I/Os are multiplexed with eight ADC channels. A dedicated internal control signal selects between the digital and analog functions

3.2 UART

RS6241 has three embedded UART controllers: UART2 is used for debugging, while UART0 and UART1, which support automatic flow control, communicate with external devices. By default, UART0 is used for system boot configuration and firmware upgrades. The UARTs are compatible with the 16550 industry standard and support baud rates up to 3 Mbps. UART features include:

- Compliant with the industry-standard 16550 specification
- DMA controller interface support
- 64-byte receive and transmit FIFOs
- Programmable fractional baud rate
- Automatic hardware flow control
- Baud-rate detection
- RS-232 support
- Clock detection to wake the system from standby

3.3 SPI

RS6241 supports a four-wire SPI interface (QSPI). QSPI supports half-duplex and full-duplex synchronous serial communication with peripherals. The interface can be configured as master or slave. Four combinations of serial-clock phase and polarity are supported. Features include:

- Clock frequency up to 96 MHz
- Motorola SPI mode
- Programmable master or slave mode
- Standard, dual, and quad SPI modes
- Programmable data width from 4 to 32 bits
- 32-word FIFO depth for both transmit and receive data
- Programmable sampling delay in master mode
- Programmable serial-clock polarity (SCPOL) and serial-clock phase (SCPH)
- Interrupt and DMA support

- Programmable instruction length, address length, and wait cycles in master mode
- Clock detection to wake the system from standby

3.4 I2C

RS6241 supports I2C0 and I2C1 communication and provides extensive pin multiplexing. I2C features include:

- Configurable as master or slave
- Standard mode (100 kbit/s)
- Fast mode (400 kbit/s)
- Fast-mode Plus (1,000 kbit/s)
- 7-bit and 10-bit addressing modes
- Multi-master systems
- 8-byte receive and transmit FIFO depth
- Programmable SDA hold time
- DMA support
- Address detection to wake the system from standby

3.5 PWM

RS6241 provides four PWM generators. Each generator outputs two PWM signals and one pair of complementary signals. Capture mode is available only on PWM channels 0, 2, 4, and 6. Capture detects rising and falling edges and calculates the capture-counter value for comparison with the match value.

PWM features include:

- Four PWM generators, eight PWM outputs, and four capture inputs (PWM0, PWM2, PWM4, and PWM6)
- Input capture and interrupt support
- Configurable dead time
- Pulse, periodic, and complementary signal outputs
- Adjustable duty cycle from 0% to 100%

3.6 Audio PWM

PWM Audio is an efficient mono Class-D audio amplifier. The digital audio interface accepts 32 kHz, 44.1 kHz, and 48 kHz sample rates and supports 16-bit and 32-bit data in all formats. The PWM Audio pins are multiplexed with PWM.

Audio PWM features include:

- Mono mode
- Configurable sample rates: 32 kHz, 44.1 kHz, and 48 kHz
- FIFO depth of 64 in mono mode
- Volume control
- Interrupt support

3.7 GPADC

The GPADC includes a 10-channel analog switch and an asynchronous 12-bit SAR ADC with single-ended inputs. Channels 0 through 7 measure external input voltages, channel 8 measures VINT, and channel 9 is dedicated to measuring VCC.

GPADC features include:

- 12-bit resolution with 10 effective bits for the SAR A/D converter
- 10-channel multiplexer with eight general channels and two VCC voltage-detection channels
- Data FIFO depth up to 32

- DMA support
- Calibration and compensation support
- Internal VREF of 1.2 V with five input-voltage attenuation ratios: 2/3, 1/2, 1/3, 1/5, and 1/9. The maximum voltage is 3.6 V
- Maximum sampling frequency: 2 MHz
- Programmable oversampling: 16, 32, 64, 128
- 4,096× oversampling on one channel
- Comparison and data interrupts
- Four operating modes: single, single-cycle, continuous, and burst
- Hardware triggers (periodic internal-timer events or WIO input events) can start conversion in standby mode. When the data falls outside the comparison window, an interrupt is generated to wake the system.

3.8 CAN-FD

The CAN controller supports CAN 2.0B, CAN FD, and TTCAN. It includes a configurable receive buffer, two transmit buffers, and four acceptance filters.

The CAN controller supports the following features:

- CAN 2.0B: Payloads up to 8 bytes and data rates up to 1 Mbit/s
- CAN FD: Payloads up to 64 bytes; the data rate is limited by the transceiver and the CAN-CTRL core clock frequency
- TTCAN: Time-triggered CAN that can be used with CAN 2.0B or CAN FD
- Configurable receive-buffer size: three buffer slots with FIFO (first-in, first-out) behavior
- Transmit buffers: one primary transmit buffer and three secondary transmit buffers
- Four acceptance filters: the acceptance code can be checked bit by bit
- Extended functions: single-shot transmission mode and listen-only mode

3.9 LIN

The LIN controller supports the following features:

- Transmission rates up to 1 MHz
- Master mode initiates message transmission and reception for different frame IDs
- Slave mode responds to and processes messages with different IDs
- Arbitration adaptation for multiple slaves
- Adjustable break-field length
- Classic and enhanced checksum modes
- Configurable frame-slot widths in master mode

4 Peripheral Interface Guidelines

4.1 GPIO Usage Precautions

Some GPIOs have special characteristics. Observe the following precautions in application designs:

- (1) PA0-PA15 and PB0-PB1 provide a WIO function and support interrupt-triggered system wake-up in low-power scenarios. Other GPIOs do not provide this function.
- (2) PA13 can place the chip in ATE test mode. This function is intended only for chip production testing and is not applicable to application development. PA13 has a weak pull-up when NRST is driven low or during initial power-on. PA13 must not be pulled low when NRST is released high or during initial power-on. PA13 can be used normally after chip startup completes.
- (3) During chip power-on or reset startup, PB0 and PB2 output high levels for several tens of microseconds.
- (4) After chip power-on or reset startup, PA0 and PA1 are weakly pulled-up inputs.
- (5) Leave unused I/Os unconnected.

4.2 Recommended Interfaces

The following interfaces are recommended for chip application designs:

- (1) PB0 and PB1 are the default UART0, used as the chip firmware programming and communication interface, and support low-power wake-up.
- (2) PB2-PB7 are the dedicated external Flash interface for applications.
- (3) PA0 is the default UART1-TX data-reporting interface in the firmware. In application designs, use PA0 preferentially for debugging or data reporting.

4.3 Entering Programming Mode

Depending on the hardware configuration, either of the following methods can be used to place the chip in programming mode.

Method 1: The host controller can control the NRST pin through an I/O

- (1) Keep VCC powered and PB0 and PB2 high.
- (2) Drive NRST low and wait 20 ms.
- (3) Drive NRST high and wait 50 ms.
- (4) Drive PB0 (or PB2) low, and then drive NRST low.
- (5) Wait 20 ms, and then drive NRST high.
- (6) Wait 50 ms, and then drive PB0 (or PB2) high. The chip and Flash have now been reset and entered programming mode.

Method 2: For systems with an externally controlled power-off mechanism

- (1) Power off the Possumic chip and Flash or the module. If the external capacitance is 10 μ F, a 10 s power-off interval is recommended. Adjust the interval according to the external capacitance and the actual discharge rate.
- (2) Drive PB0 (or PB2) low.
- (3) Power the Possumic chip and Flash or the module back on, wait 50 ms, and drive PB0 (or PB2) high. The chip and Flash have now been reset and entered programming mode.

4.4 Programming Interfaces

The chip provides UART, I2C, and SPI firmware programming interfaces. The communication interfaces are configured by polling. The default programming interface configuration is as follows:

- (1) UART uses PB0 and PB1 for the UART0 interface

- (2) I2C uses PA4 and PA5 for the I2C0 interface
- (3) SPI uses PA0-PA3 in serial simplex mode

After the chip enters forced programming mode, the UART, I2C, and SPI interfaces poll for handshake communication. Firmware programming begins after a successful handshake.

5 Power-On Sequence

5.1 Power-On Sequence

The chip has no special power-on sequence requirements for its power rails. However, observe the following precautions for different system configurations:

- (1) **If an SPI NOR Flash is used as the boot device and NRST is pulled up to VDDIO by default without external control:**

After the chip powers on and NRST is released (the voltage rises to a high level), the chip starts immediately and attempts to read firmware from the SPI NOR Flash. The SPI NOR Flash, however, can be accessed only after a specified interval following VCC power-on. In typical designs, VDDIO supplies the SPI NOR Flash. Therefore, add an RC delay circuit around NRST so that, during power-on, NRST is released only after the SPI NOR Flash has powered on and the required delay has elapsed.

Calculate the RC delay-circuit parameters according to the power-on timing requirements in the SPI NOR Flash datasheet.

- (2) **If an SPI NOR Flash is used as the boot device and NRST is controlled by an external host controller, reset and start the chip by controlling NRST after power-on is complete, the external host controller is ready, and the SPI NOR Flash has powered on.**

6 Package Description

6.1 Package Outline

RS6241 uses an FCCSP package with a bump diameter of 350 μm , a pitch of 650 μm , a standoff of 250 μm , and dimensions of 6.7 $\times$ 6.7 $\times$ 0.87 mm³.

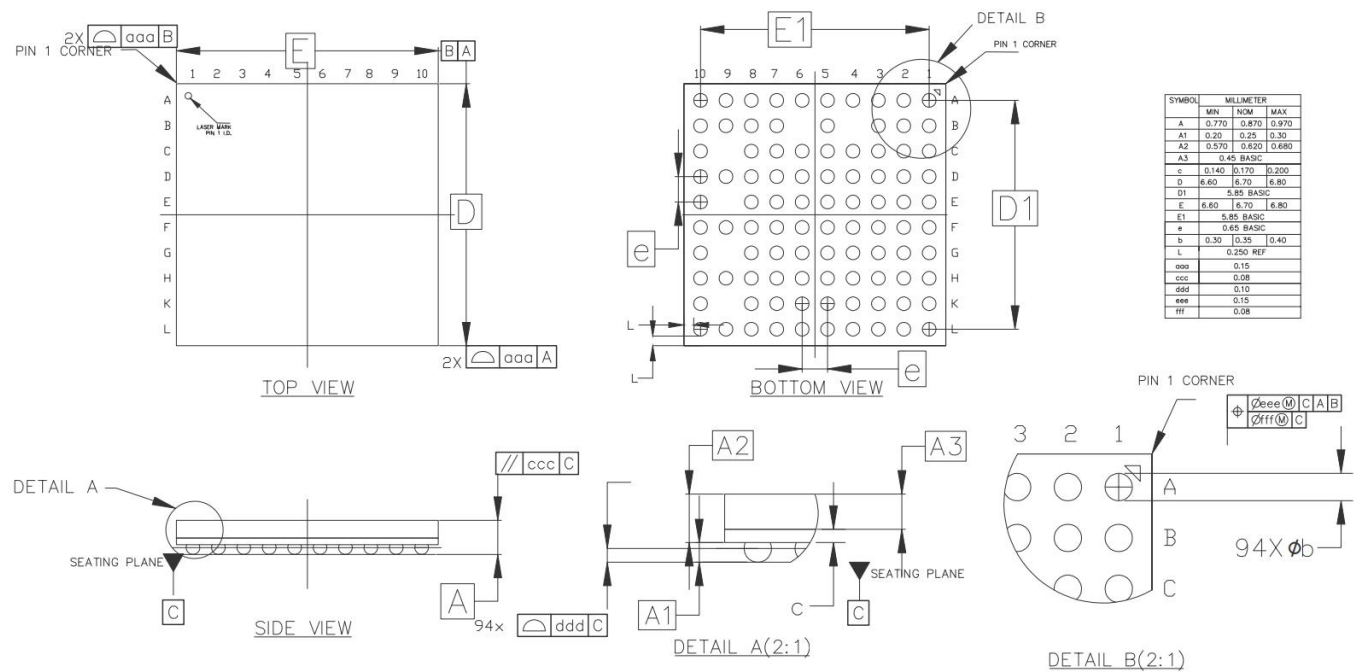


Figure 6-1 Package Outline

6.2 PCB Land Pattern

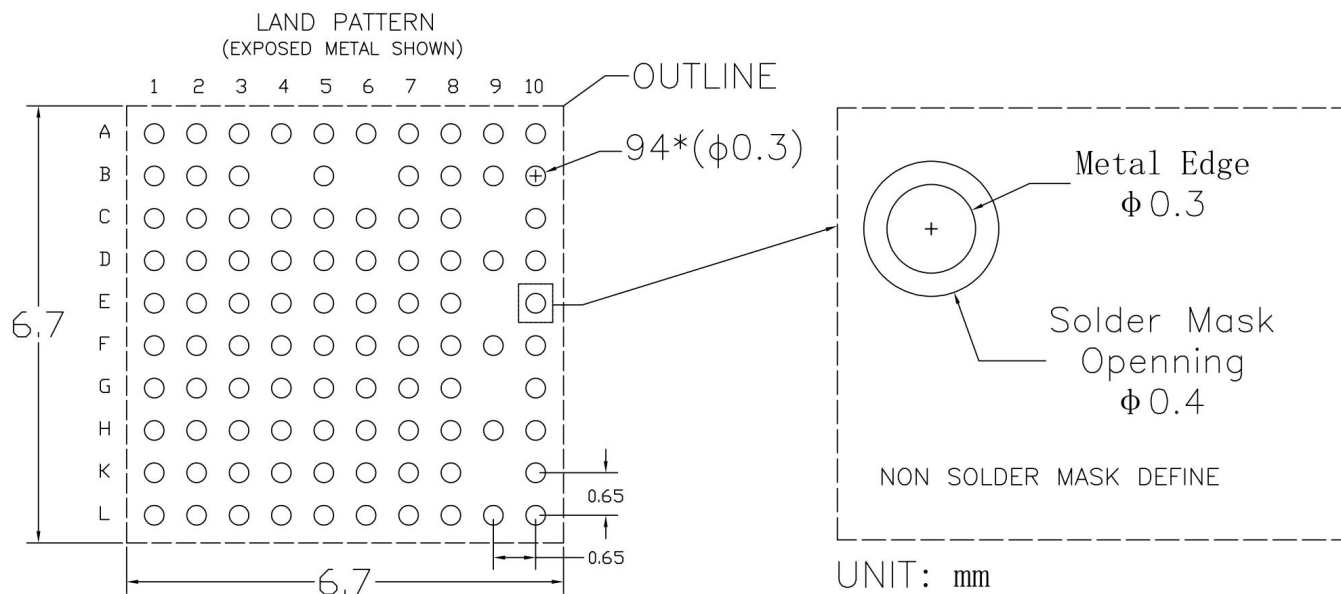


Figure 6-2 PCB Land Pattern

7 Reference Schematic

7.1 Precautions

- **Power Supply**

- **OPTION A:** Compared with supplying V13 from the internal RS6241 LDO, using an external DC-DC supply reduces power consumption. If power consumption is not critical, this circuit can be omitted to reduce BOM cost.
- VCC and VDDIO are supplied externally. If VCC is below 3.6 V, VCC and VDDIO can be tied together and supplied from one power source.

- **Bluetooth**

- **OPTION B:** "32.768 kHz" is used by the Bluetooth subsystem. If Bluetooth is not required, this circuit can be omitted.
- **OPTION C:** "BLE ANTENNA" is used by the Bluetooth subsystem. If Bluetooth is not required, this circuit can be omitted.

- **External Connections**

The host controller can control the radar through UART0, I2C0, or SPI0. UART0 (PB0/PB1) is used by default.

PA6/PA11 are used as radar interrupt signals.

If VCC has no power switch, control NRST from the host MCU.

If over-the-air (OTA) firmware updates are required, route out PB0 to control the chip entering programming mode. See Section 4.3.

To facilitate radar performance analysis, provide PA0, PB0, and PB1 as test points. These three signals are used for noise-floor testing.

- **OPTION D:** Connect via I2C0 (PA4/PA5), radar interrupt signal (PA6/PA11)
- **OPTION E:** Connect via SPI0 (PA0/PA1/PA2/PA3), radar interrupt signal (PA6/PA11)

- **PSRAM**

- **OPTION F:** PSRAM is used for memory expansion. If data-intensive functions such as posture detection or high-performance human-presence sensing are not required, this circuit can be omitted. Keeping it is recommended.

- **Leave unused pins unconnected.**

When designing the circuit, carefully review the description of each circuit module in the following table:

Table 7-1 Circuit Module Descriptions

Module	Component	Description
RS6241	C18,C19,C20,C21,C22,C23	Chip decoupling capacitor (X5R, $\pm 10\%$ tolerance, 6.3 V rating). Do not change the capacitance value.
	R1,C2	NRST power-on RC delay to ensure the Flash is ready before NRST is released. Set the parameters according to the Flash power-on sequence. See Section 5.1, Power-On Sequence, in 'RS6240_Hardware_Design_Guide_EN.pdf'.
RADAR ANTENNA	E3,E4,E5,E6,E7,E1	Use 50-ohm impedance-controlled routing
DCXO 32M	U6,C16,C17	Crystal specification: 32 MHz, 12 pF, ± 10 ppm. C16 and C17 (C0G, $\pm 1\%$ tolerance, 6.3 V rating) are matching capacitors. The chip has built-in capacitors, so these components are NC by default.
32.768 kHz [Option B]	U7,C14,C15	Low-frequency crystal: 32.768 kHz, ± 20 ppm. C14 and C15 (C0G, $\pm 1\%$ tolerance, 6.3 V rating) are matching capacitors and must not be NC; select the values according to the crystal. This crystal provides accurate timing for the Bluetooth system in standby. If Bluetooth is not used, this circuit can be omitted.
BLE ANTENNA [Option C]	R6,L3,L4,E1	Bluetooth antenna and matching circuit. Modify the matching circuit for the selected antenna type, and route the antenna feed with 50-ohm impedance control. If Bluetooth is not used, omit the circuit and leave the signals unconnected.
SPI NOR	R3,R4,R5,U3	R3, R4, and R5 are pull-up resistors ($\pm 5\%$ tolerance). U3 is the Flash; select it according to 'Key_BOM_Selection_Guide_EN.pdf'.
PSRAM [Option F]	R2,U4	R2 is a pull-up resistor ($\pm 5\%$ tolerance). U4 is an APS1604M-3SQR-ZR PSRAM that shares the SPI signals with the Flash through time-division multiplexing. PSRAM is used for memory expansion. If data-intensive functions such as posture detection or high-performance human-presence sensing are not required, this circuit can be omitted; however, keeping the footprint without populating it is recommended.
DC-DC [Option A]	Selection Notes	Compared with supplying V13 from the internal RS6240 LDO, using an external DC-DC supply reduces power consumption. If power consumption is not critical, this circuit can be omitted to reduce BOM cost.
	L1,C3,C8,C11	Typical JW5712A application circuit L1: DC-DC matching inductor, $DCR \leq 300 \text{ m}\Omega$, rated current $\geq 300 \text{ mA}$, such as LQM18PN2R2MFRL C: Matching capacitor (X5R, $\pm 10\%$ tolerance, 6.3 V rating)
	U2	JW5712A low-power DC-DC converter. Replacing it with another model is not recommended. A replacement must have a quiescent current below 600 nA, an operating current above 400 mA, and a 1.3 V output. For other models, see 'Key_BOM_Selection_Guide_EN.pdf'.
	L2	DC-DC output filter ferrite bead, $1 \text{ k}\Omega$ at 100 MHz, $DCR \leq 1.1 \text{ }\Omega$, rated current $\geq 250 \text{ mA}$, such as BLM15HG102SN1D.
	C5,C6,C7	DC-DC output filter energy-storage capacitor (X5R, $\pm 20\%$ tolerance, 6.3 V rating). The matching capacitance varies with the supply type in low-power applications: ➤ Batteries with high internal resistance, such as coin cells and dry cells: Add 200 μF at the battery, reserve two additional capacitor footprints, and populate C5, C6, and C7 with 47 μF each. If layout space is insufficient, the capacitance can be reduced as appropriate. Verify performance according to Section 3.2, DC-DC Output Performance Measurement.

		➤ Batteries with low internal resistance, such as lithium batteries or always-powered supplies: Reserve two additional capacitor footprints at the battery, populate C5 and C6 with 10 μF each, and leave C7 NC.
Connect via UART0 [Default]		➤ The host controller can control the radar through UART0, I2C0, or SPI0; UART0 is used by default. To facilitate radar performance analysis, provide PA0, PB0, and PB1 as test points. These three signals are used during PCB performance analysis. ➤ VCC and VDDIO are supplied externally. If VCC is below 3.6 V, VCC and VDDIO can be tied together and supplied from one power source. ➤ PA6/PA11 are typically used as interrupt-trigger signals to wake the host. ➤ If online firmware updates are required, route out PB0 to control the chip entering programming mode. ➤ If VCC has no power switch, connect NRST to the host controller to support recovery resets or online firmware updates.
Connect via I2C0 [Option D]		
Connect via SPI0 [Option E]		

8 PCB Layout Guidelines

8.1 Component Placement and Routing

A four-layer PCB is recommended for radar hardware applications. The following layout guidelines assume a four-layer PCB.

8.1.1 PCB Layout Requirements

- For top-layer component placement, place the VRF, V13, VCC, VDDIO, VEXT, and VDIO capacitors close to the chip and their corresponding signal pins, within 2 mm.

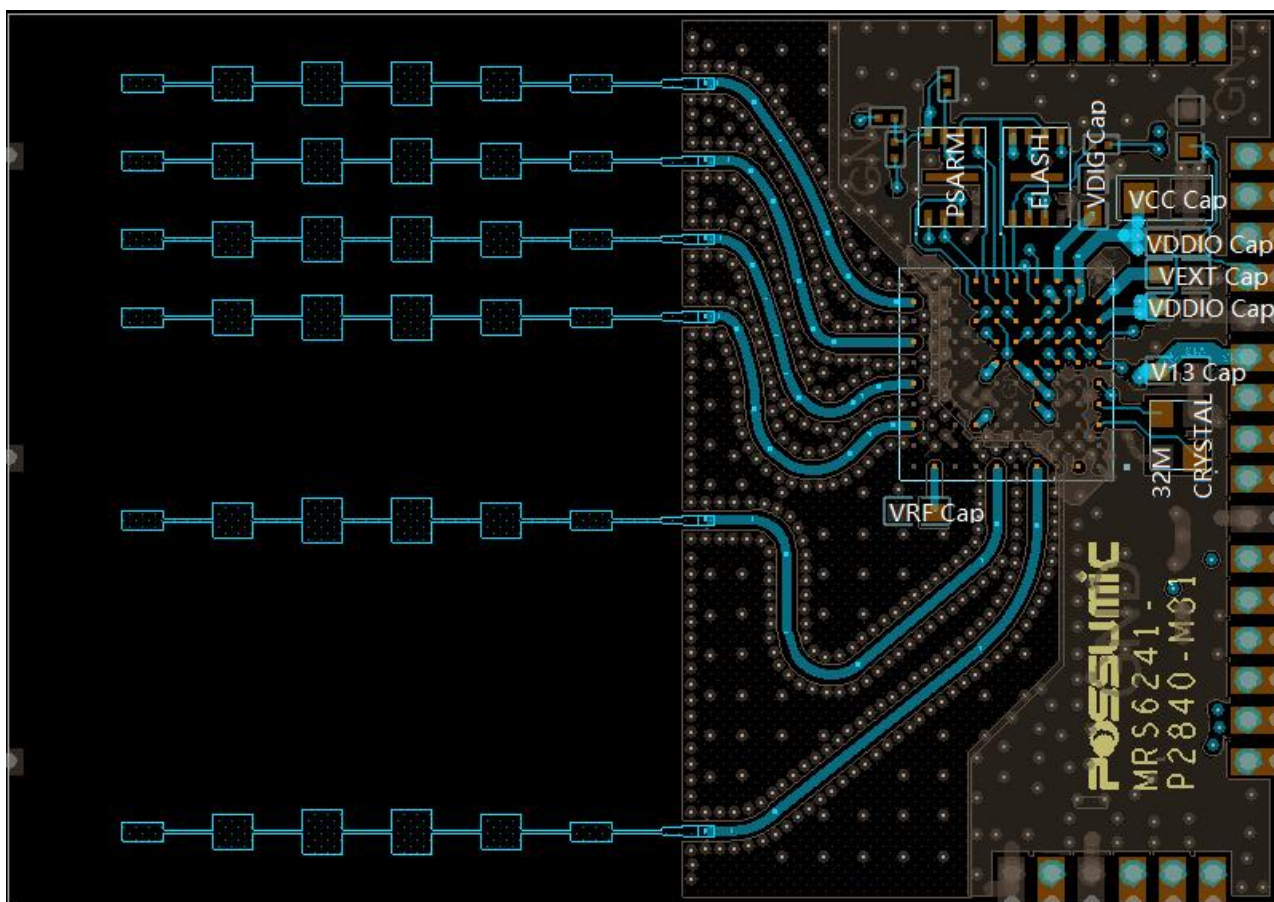


Figure 8-1 PCB Top-Layer Layout (Radar-Chip Side)

- For a single-sided component layout, use a solid GND plane on the second layer, as shown below:

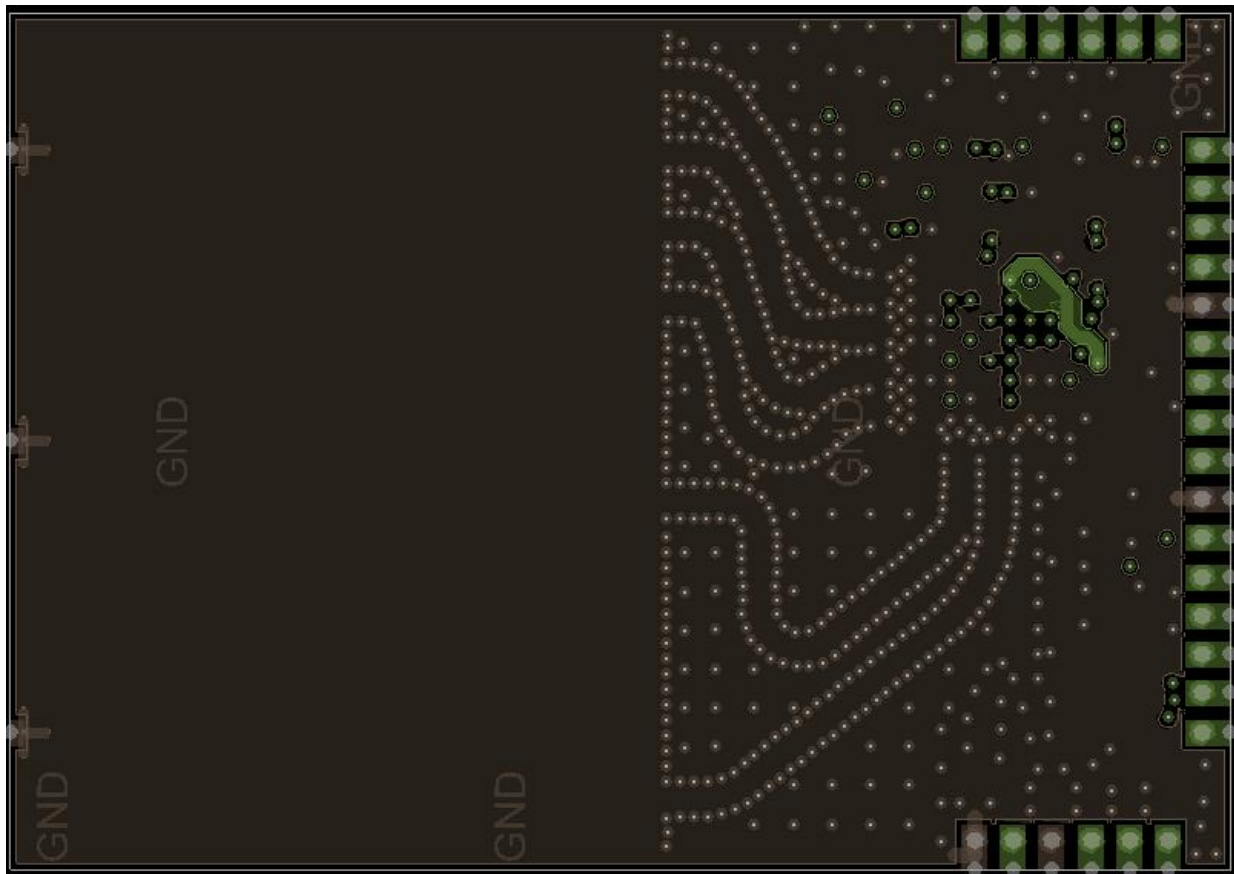


Figure 8-2 PCB Second-Layer GND Plane

- Route power and signals primarily on the top, third, and bottom layers.
- Power trace width requirements: VCC, VEXT, and V13 greater than 0.5 mm; VDDIO, VDIG, and VRF greater than 0.25 mm.
- V13 is sensitive. Do not overlap its routing with signal traces or VDDIO on adjacent layers.

8.1.2 32 MHz Clock Layout

- Top-layer layout example for the 32 MHz crystal clock:

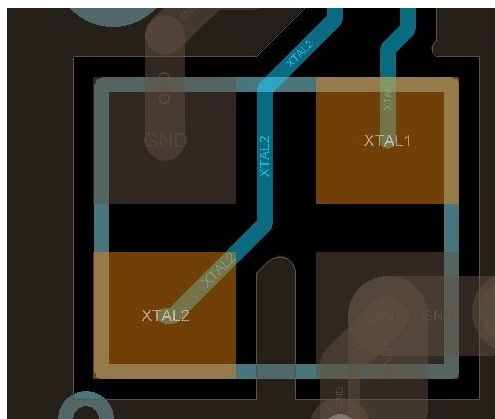


Figure 8-3 Clock Layout

- 32 MHz crystal-clock layout requirements:
 - Connect the crystal GND vias to the second-layer GND plane.

8.1.3 External SPI Layout

When using SPI to communicate with an external host, observe the following requirements:

- Control single-ended trace impedance to 50 ohms ($\pm 10\%$).
- Keep the clock-to-data length mismatch below 5 mm.
- Keep traces shorter than 10 cm.
- Keep spacing between signal traces at least twice the trace width.

8.2 Layout Example (Without DC-DC/Bluetooth)

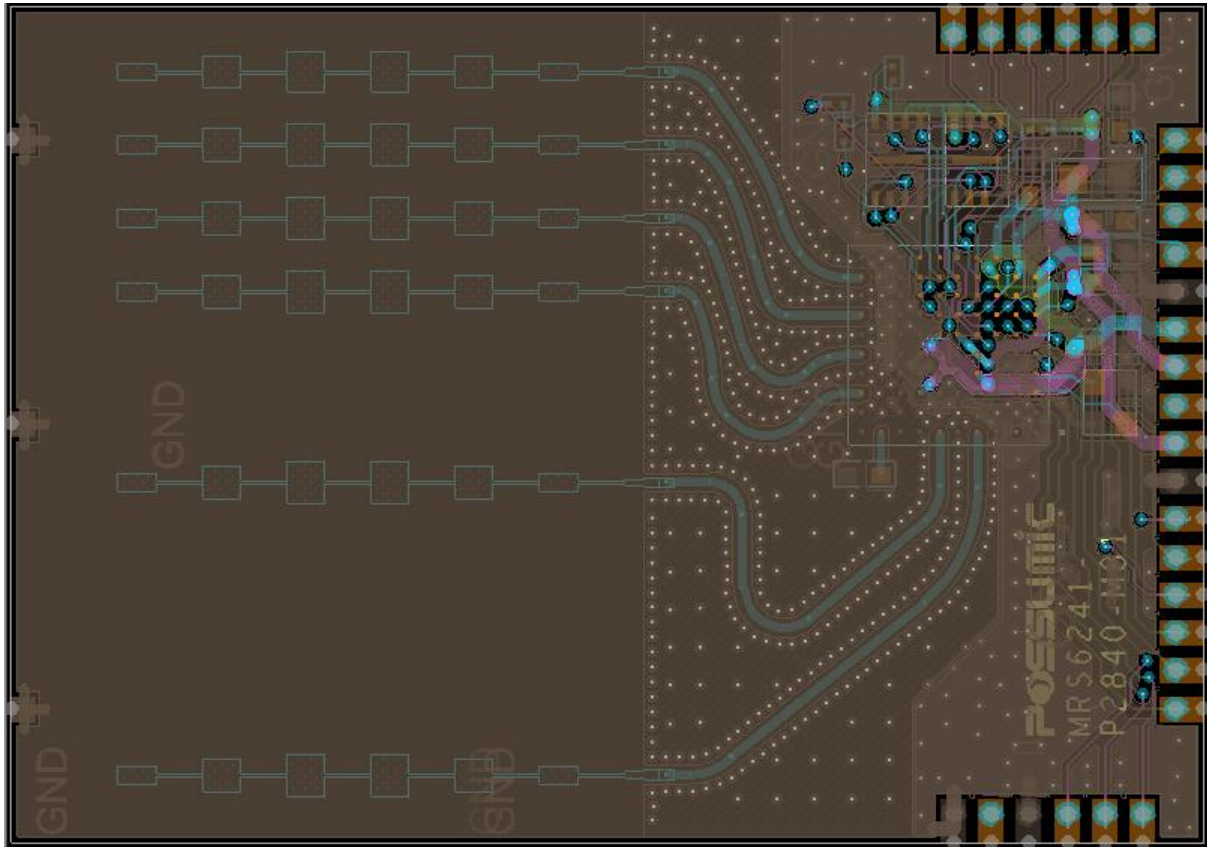


Figure 8-4 Overview

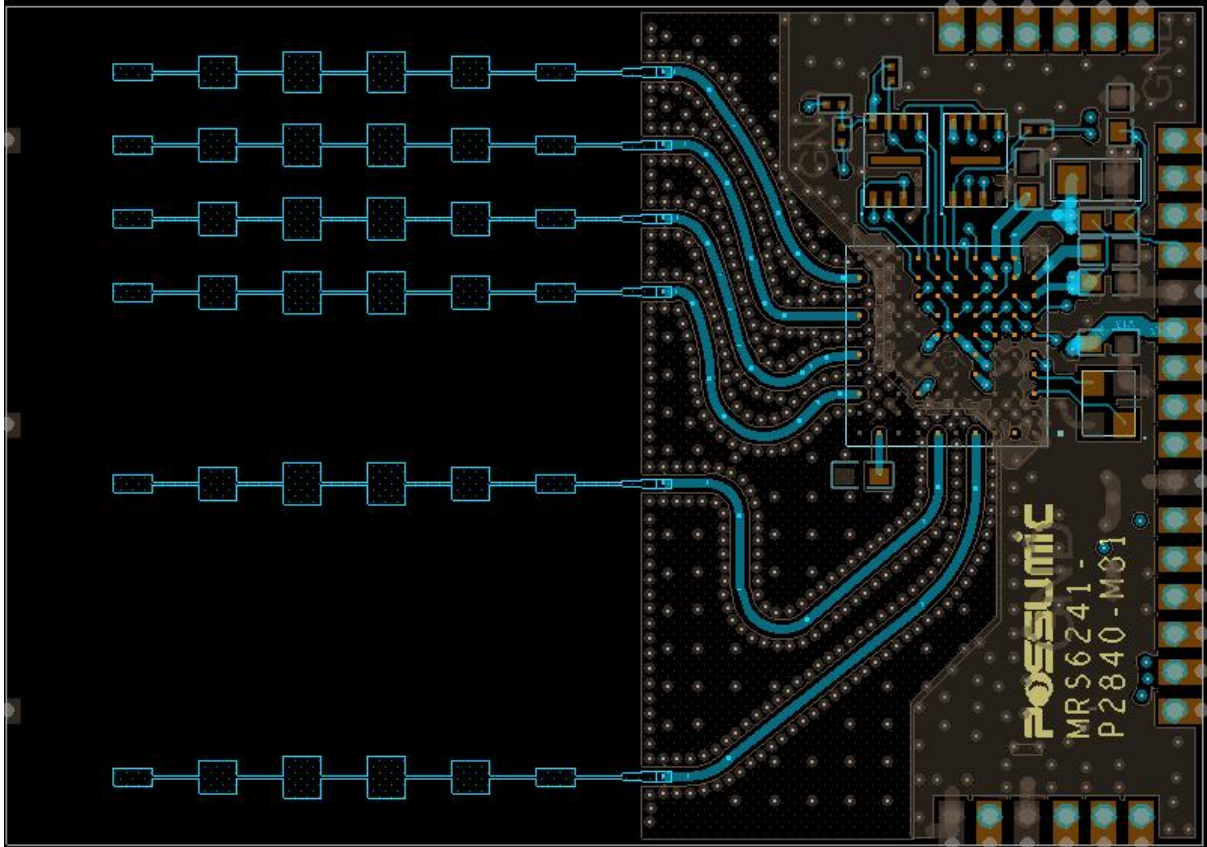


Figure 8-5 Top Layer

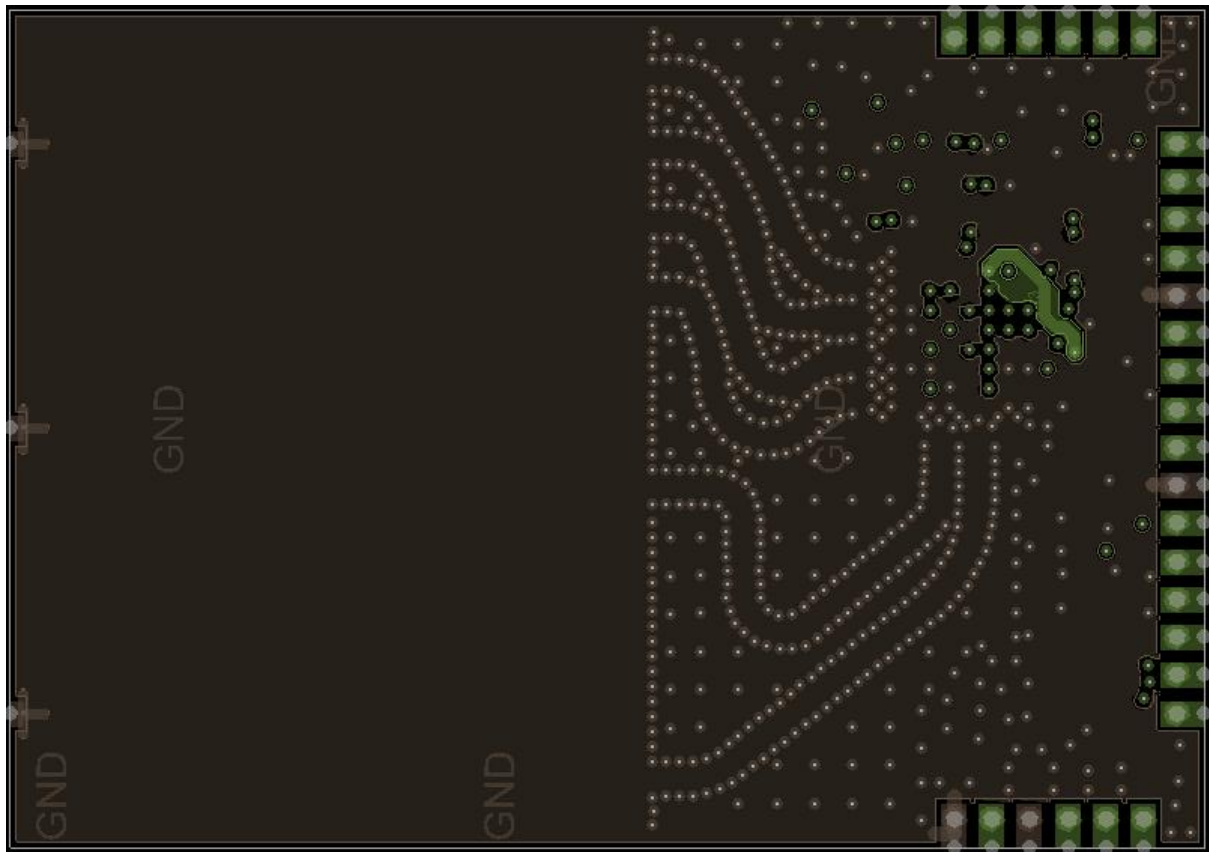


Figure 8-6 Second Layer

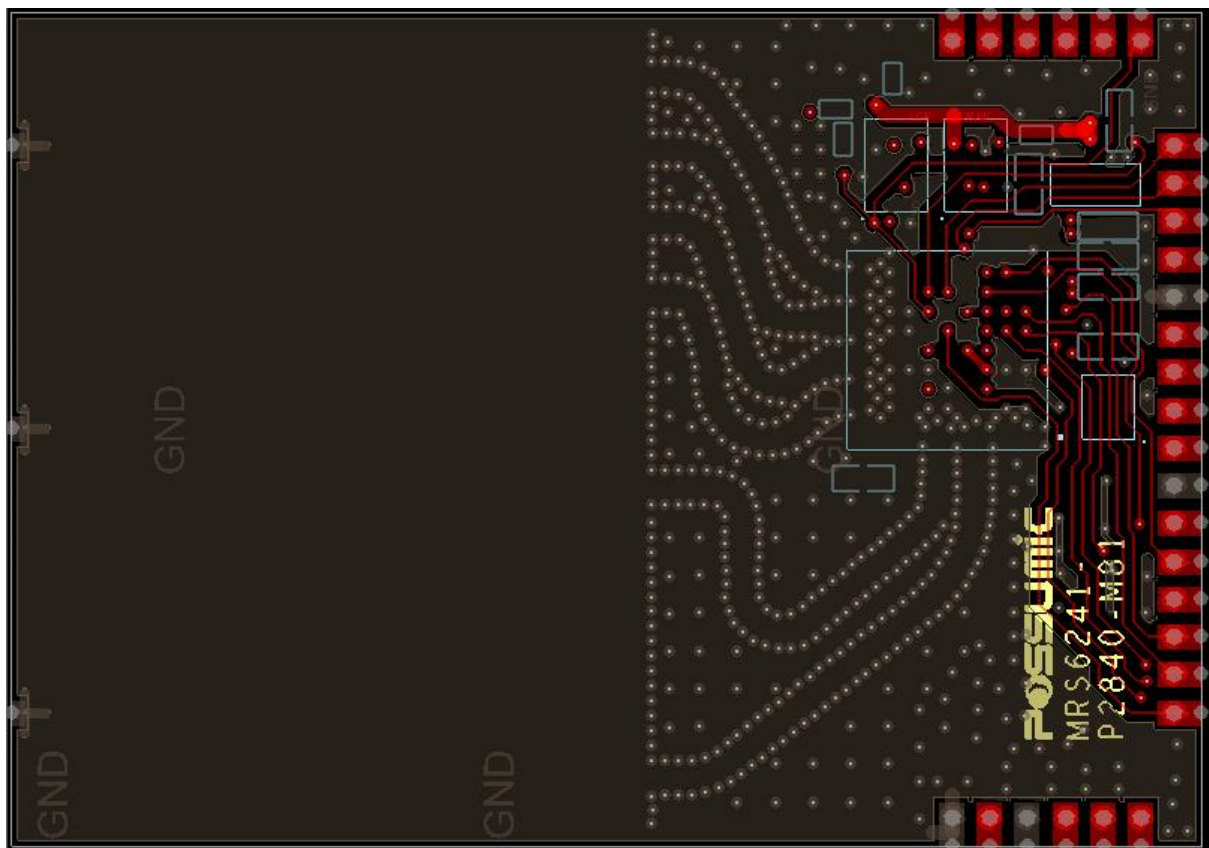


Figure 8-7 Third Layer

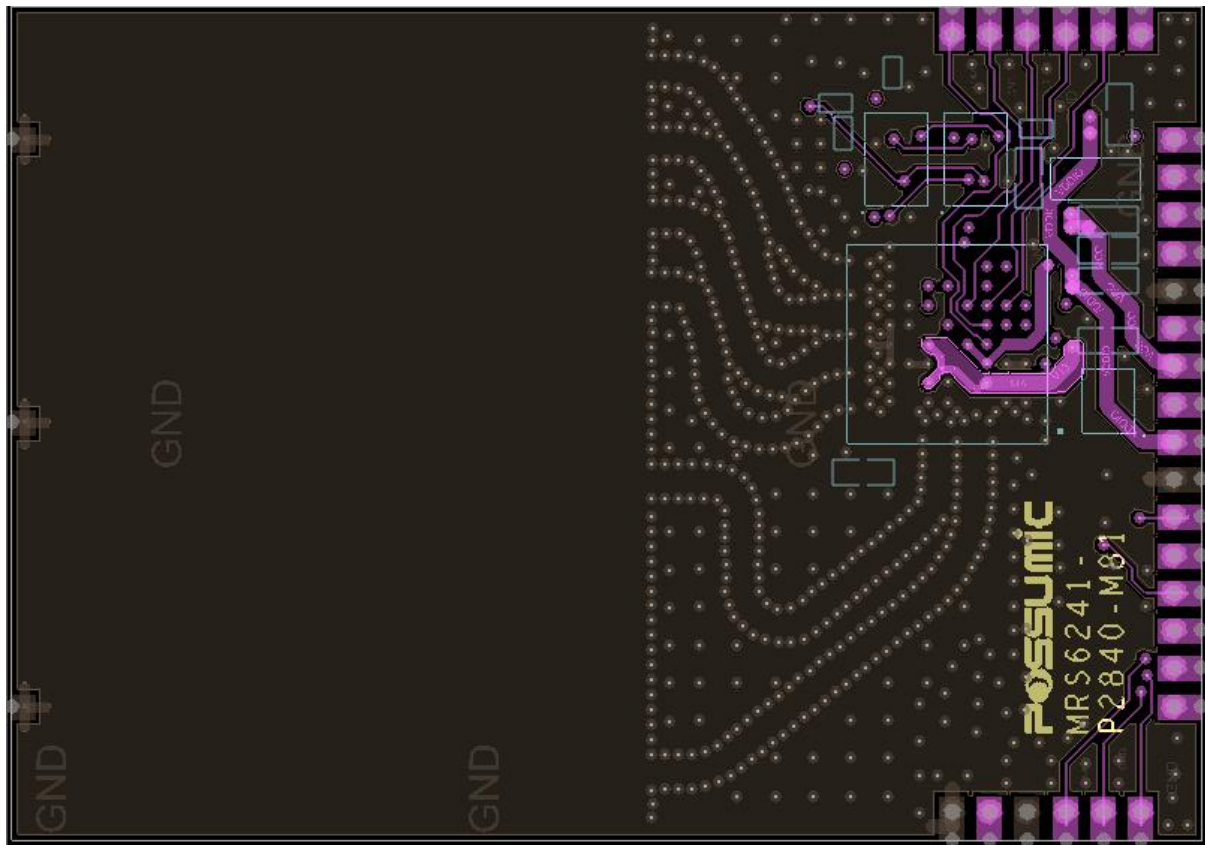


Figure 8-8 Bottom Layer

9 PCB Performance Analysis

9.1 Noise-Floor Measurement

A poor PCB design can introduce excessive noise into the radar system and degrade ranging performance. The following noise-measurement method helps evaluate PCB design quality.

The test requires PA0 (UART1-TX), PB0 (UART0-TX), and PB1 (UART0-RX). Ensure that these three signals are accessible in the PCB design.

Test preparation:

- PC tool: RadarAnalysisTool
- PCB under test (program the dedicated test firmware radar_analysis_rs6240_p2512.img)
- UART adapter board (connected to UART0-TX, UART0-RX, and UART1-TX on the PCB)
- Test environment: Ensure that only the radar under test is operating nearby. Other operating radars will cause interference.

Note: The PC tool and dedicated test firmware are available in SDK\Software_Kit\04_image.

9.1.1 PC Tool Operation

Step 1:

Power on the PCB, connect the two UARTs described above to the PC, and open the PC tool.

- (1) Click 'Platform'.
- (2) Select the MRS6240-P2512 platform corresponding to the hardware.
- (3) Connect the UARTs. For the control port, select the COM port corresponding to UART0 and set the baud rate to 115200. For the data port, select the COM port corresponding to UART1 and set the baud rate to 1000000.
- (4) Click Connect Device.

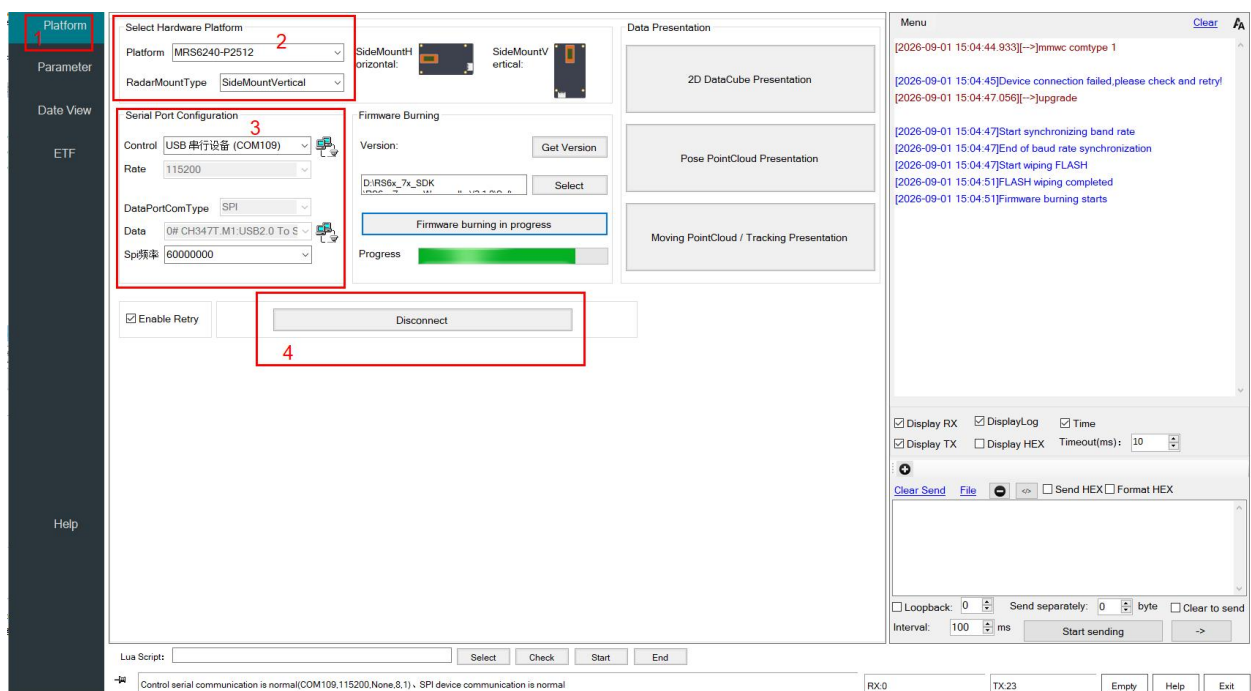


Figure 9-1 PCB Performance Test - Step 1



Figure 9-3 PCB Performance Test - Step 3

9.1.2 Acceptance Criteria

A well-designed hardware platform has the following measured noise characteristics:

- The overall amplitude is centered at approximately -76 dB.
- The amplitude curve changes smoothly without prominent peaks.

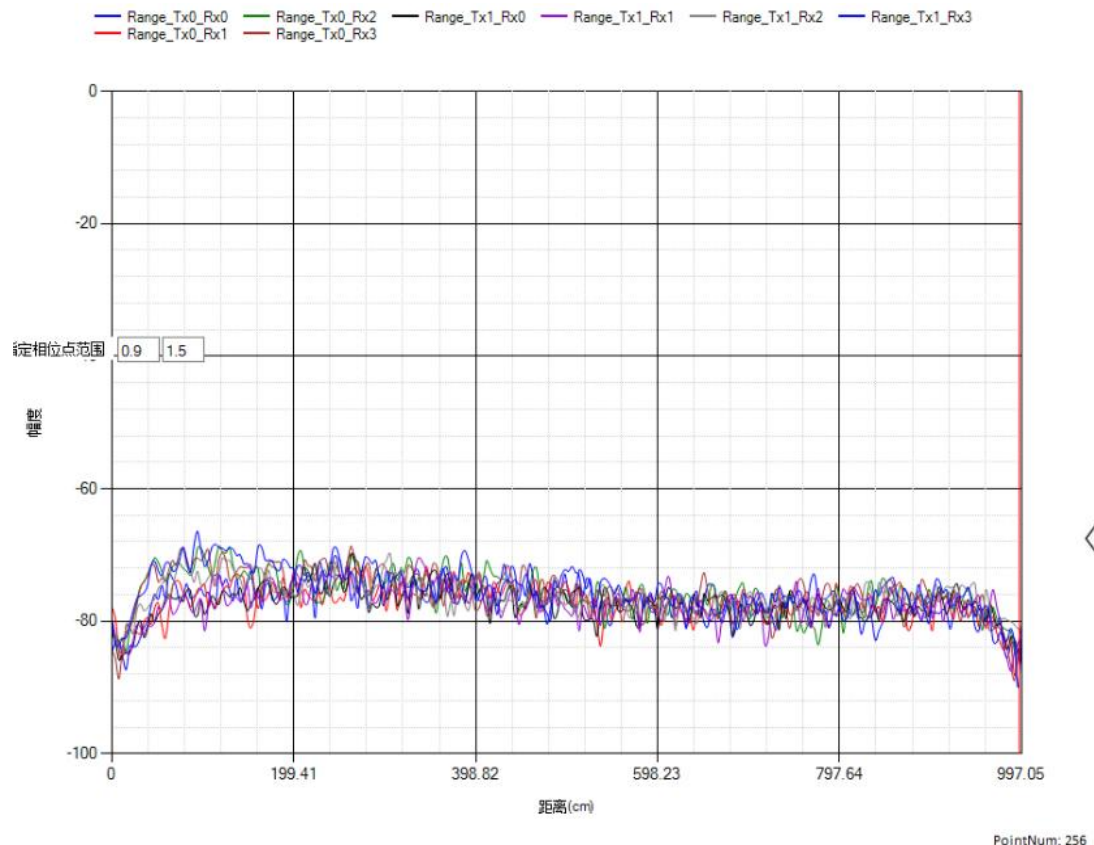


Figure 9-4 Normal Plot

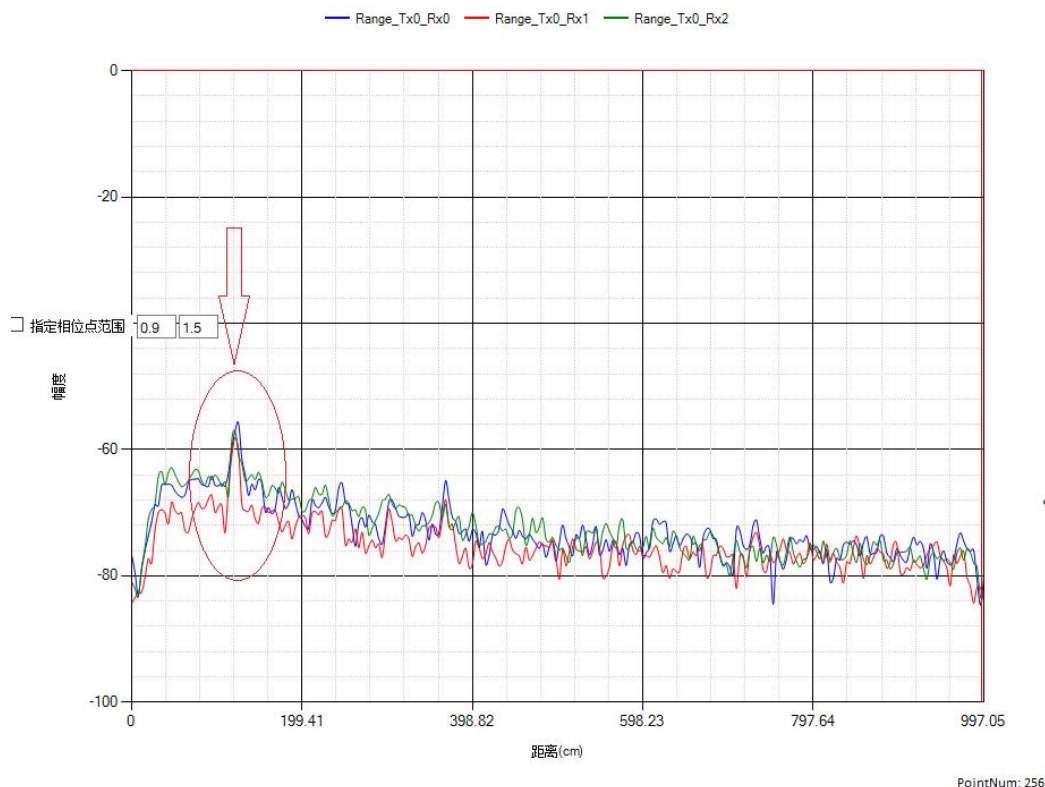


Figure 9-5 Abnormal Plot (RS6130 Affected by DC-DC Interference)

9.2 DC-DC Output Performance Measurement

If the design uses a DC-DC converter, measure its output performance to ensure that it meets the radar V13 power-input requirements.

- During radar operation, mmWave signals are transmitted periodically to sense the environment, causing periodic current draw from V13. A brief V13 voltage drop is unavoidable when transmission starts. Minimize the DCR in the path from the DC-DC output to the chip V13 pin so that the drop is less than 100 mV and the post-drop voltage remains above 1.2 V.

DC-DC switching noise can couple into the V13 network and create supply ripple (excluding the voltage drop at transmission startup described above). Excessive ripple interferes with the radar RF circuitry. Place suitable capacitors on the V13 power network so that the measured V13 ripple is less than 50 mV peak-to-peak.

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